The Paradox of Radii: Convection of Meanings, Ontology of Analogies, and Fundamental Connectivity

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Abstract

This essay explores the heuristic potential of scientific analogies through the lens of a specific paradox arising from the rubber sheet model of General Relativity. The "paradox of radii"—concerning the relationship between a massive body and its gravitational well—serves as a starting point for examining how analogies function not merely as pedagogical simplifications but as genuine cognitive tools that reveal deep structural connections between distinct ontological layers. We argue that analogies create entangled pairs between domains (mechanical, geometric, semantic), enabling a circulation of meanings across these layers. This process is governed by a proposed Law of Conservation of Meanings, which posits that meaning, broadly construed as information, is neither created nor destroyed but transforms across different modes of existence. The framework is applied to fundamental problems including the black hole information paradox, the hard problem of consciousness, and artificial intelligence, suggesting that ontological entanglement and meaning conservation offer productive avenues for theoretical unification. We conclude that the study of analogy provides a key to understanding the multi-layered yet fundamentally interconnected structure of reality, or "Superreality."

Keywords: analogy, General Relativity, ontological entanglement, conservation of meaning, Superreality, black hole information paradox, philosophy of science, scientific modeling, consciousness, rubber sheet analogy

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Prologue: From Paradox to Inquiry

Starting Point. We noticed a popular analogy—the rubber sheet—used to illustrate gravity in General Relativity: a massive body (a ball) depresses the surface, forming a

"funnel." We wondered: if two bodies create their own funnels, does a "ridge" emerge between them—an analogue of an energy barrier? Is this confirmed by calculations?

What We Found. The analogy is useful but imprecise: for ordinary bodies (planets, comets), no external "ridge" exists—it lies deep inside, beneath the surface. However, the model itself revealed a specific incongruity: the depression is often depicted as wider than the ball, yet the ball rests *on* the rubber, not inside it. A "paradox of radii" emerges: is the radius of the depression (the region of indentation) smaller or larger than the body's radius? In any case, the visual image does not align with physical logic and *remains elusive within the model's own framework.*

Initial Reaction. Analysis grounded in GR decisively rejects this paradox as an artifact of the analogy: the theory has no concept of a "depression radius"; it has the gravitational radius r_s, and for ordinary bodies this is much smaller than the physical radius—no contradiction exists.

But. Dismissing such an unusual conclusion-phenomenon would be too hasty. What if this is not merely a model's flaw but an indicator of deeper connections? What if "paradoxical" consequences deserve exploration rather than rejection?

The Central Question. Can such paradoxes be heuristically valuable? And can analogies, rooted in real phenomena, serve as tools for thought experiments that lead to new insights?

Part I. Analogy and the "Arena"

Why Analogies Matter

General Relativity describes gravity as a geometric property of spacetime. This is a profoundly abstract idea with no direct counterpart in everyday experience. Analogies (the rubber sheet, magnetic field, fluid flow) act as "translators": they render the complex mathematical language of curved geometry into vivid imagery with which we can mentally experiment. Each analogy inevitably simplifies, but it is precisely this simplification that allows us to grasp the core idea—the curved *arena.*

What Is the "Arena" (A Metaphor)

In the context of our discussion, the **arena** is a metaphor for the "container" within which motion unfolds. In GR, this container is spacetime itself, whose geometry is curved by the presence of mass-energy. It is important to distinguish the arena from other concepts:

- **Not ether:** ether was conceived as an elastic medium propagating waves; the arena is not a substance—it is pure geometry.

- **Not a field:** a field (e.g., electromagnetic) exists *in* space and exerts forces; the arena *is* space itself, and bodies move inertially, following its curvature.
- **Not merely space:** Newtonian space was an empty, immutable container; the GR arena is active—its geometry changes under the influence of masses and, in turn, dictates how those masses move.
- **Geometry as essence:** the arena is geometry understood as the set of rules for measuring distances, times, and angles. The curvature of the arena means these rules cease to be Euclidean and become dependent on place and time.

Three Illustrative Analogies

1. **The Rubber Arena** (depression under a ball)—a pedagogical device visualizing that mass curves the geometry of space.
2. **Magnet and Iron Filings**—the magnet, through its field, organizes the filings along curved lines; a vivid illustration of how the "arena" directs motion.
3. **Fluid Flow**—the flow itself is the "arena" (an analogue of spacetime); its inhomogeneities determine the trajectories of particles moving inertially with the current.

The Mathematical Core of GR

The curvature of the "arena" begins where the metric tensor (the rules for measuring distances and time) ceases to be constant and starts depending on coordinates. This change in the metric, linked to the presence of mass-energy via the Einstein field equations, is the quantitative expression of spacetime curvature.

Part II. The Nature of Analogy

1. Analogy as Simplification (Principled Limitation)

Every analogy is a reduction, transferring only certain features of a complex phenomenon onto a familiar image. It inevitably "limps": it has boundaries of applicability beyond which distortions begin.

Our experience: The dialogue about the rubber sheet demonstrated this brilliantly. Attempting to apply the analogy literally led to the question of the relationship between the radii of the "depression" and the ball—a question that exposed an inconsistency in the model itself, not in the theory. This is a classic case where an analogy "breaks" if asked to do too much.

2. Analogy as a Cognitive Tool (Relevance)

Despite simplification, an analogy is valuable because it *relevantly* captures the key essence of the phenomenon. It is this relevant part that allows us, while aware of the model's limits, to experiment mentally and gain insight into the nature of the phenomenon itself.

Our experience: In the three documented analogies (rubber, magnet, flow), the relevant part is the idea of **"curvature of the arena."** Despite all differences, they each illustrate precisely this: something (mass, field, flow inhomogeneity) alters the geometry of the arena, and this altered geometry dictates to objects (test bodies, wood chips, filings) how to move inertially.

3. Thought Experiment: Reduction to Significant Elements

Within each analogy, to preserve its relevance, we consciously isolate only those entities directly involved in illustrating the phenomenon. Everything else is excluded as irrelevant.

- In the rubber analogy: only the elastic surface (arena) and heavy balls (masses) are significant.
- In the magnet analogy: only the magnet (field source) and filings (test bodies) are significant.
- In the flow analogy: only the flow itself (arena) and light particles (test bodies) are significant.

This principle of reduction allows the analogy to be used as a tool for thought experiments, undistracted by insignificant details.

> **Note (on the nature of entities).** The quotation marks around "entities" are placed deliberately. In a thought experiment, the arena is not physical rubber but an *image* of geometry; balls are not merely metal but an *image* of mass; filings are an *image* of test bodies. Moreover, entities with no material analogue at all may appear here: ideas ("geodesic"), concepts ("curvature"), imaginary constructs ("event horizon"), hypotheses. They are all equal tools of thought.

4. Analogy as a Self-Contained World

Each analogy creates an **internally consistent world**, populated only by those entities necessary to illustrate the phenomenon. By operating with the tools of this "subordinate" world, we can grasp—albeit metaphorically—properties of phenomena belonging to a more general reality (gravity, black holes, spacetime geometry).

The more relevant analogies we consider, the denser the network of our representations becomes. Each new analogy adds a facet not captured by the previous ones. In the limit, an infinite set of consistent analogies (if possible) would provide a complete description of the phenomenon itself—the phenomenon would, as it were, show through the aggregate of its models. Of course, an infinite set is unattainable in practice, but the very *tendency toward completeness through a multiplicity of models* is an important heuristic principle.

5. "Lawful Residents" of the Analogy-World

In each such world, there exist its own **"lawful residents"**—objects that fully obey its internal rules and allow modeling the phenomenon without violating consistency.

- In the flow analogy: the lawful resident is the water itself (a drop, a stream), wholly determined by the flow's geometry. Rigid objects (a boat, a wood chip) are already "unlawful residents": they respond less to the arena's geometry and more to external forces.
- In the rubber analogy: lawful residents are heavy balls (masses) and light test bodies rolling on the surface.
- In the magnet analogy: lawful residents are the magnet and filings aligning strictly along field lines.

The choice of objects is dictated by the structure of the analogy (theory) itself and defines its limits of applicability. Introducing "unlawful residents" leads to paradoxes and requires moving beyond the given model.

6. Productivity of "Paradoxical" Consequences

Key Insight. Internal "paradoxes" of an analogy, arising from its literal (deliberately expanded) reading, can serve not as evidence of its inadequacy but as **indicators pointing to the existence of another level of description** where these paradoxes are resolved.

- **Flow example:** For a water droplet, the behavior of a boat (moving upstream, maintaining its shape) seems miraculous. But this "miracle" is explained by transitioning to the world of solid-state physics.
- **Returning to the "paradox of radii":** The paradox (depression smaller than the ball?) appears absurd within the rubber analogy. Yet developing this idea leads to real questions in GR: about the relationship between the gravitational radius and the size of a body, about where the "surface" of a black hole lies. What is absurd in the model acquires precise meaning in the theory (r_s can be either less than or greater than R).

7. The Cognitive Power of the "Impossible"

The ability to mentally violate laws established within a given model (walking on clouds, passing through walls, getting 2+2=5) is not a pathology of thought but one of its most important tools.

- **Mathematics:** abandoning the parallel postulate gave rise to Lobachevskian geometry; admitting imaginary numbers provided a powerful tool for physics.
- **Science of illusions:** an optical illusion leads to investigating the mechanisms of perception; an induced hallucination leads to neurophysiology.
- **Our case:** the "paradox of radii" is such a mental walk on clouds. It allowed us to probe the boundary of the rubber analogy's applicability and arrive at the real relationship between r_s and R in GR.

We are endowed with imagination not to escape reality, but so that by mentally "breaking" one picture of the world, we can discover the contours of another, more general one.

8. The Ontological Status of Analogies

Analogies in scientific cognition are not arbitrary hallucinations. Their foundation lies in **real phenomena**, rooted in their own ontological layer (physical, chemical, biological, mathematical).

- A rubber sheet with balls is a real mechanical interaction.
- A magnet with iron filings is a real manifestation of a magnetic field.
- Fluid flow is a real hydrodynamic phenomenon.

Precisely because analogies are "not pulled out of thin air" but have an empirical basis, we are entitled to expect that their careful study, and even the "paradoxes" discovered within them, can lead not to empty speculation but to quite academic, sometimes empirically testable conclusions. An analogy is a **connection between two realities**, each authentic in its own way.

9. Analogy as an Imprint: Polymorphism of Resemblances

Analogy is not merely a comparison but an **imprint**. An imprint of a real phenomenon, an imprint of an imprint, and so on, ad infinitum. But it is crucial to understand: we are not speaking only of geometric or spatial "form." An imprint can be digital (a file

copy), semantic (a paraphrase of a thought), meaningful (an interpretation of a text)—the polymorphism of formats here is unlimited.

In the limit, the world can be represented as an infinite hierarchy of resemblances, continuously generating new levels of representation. Each imprint is not accidental. It carries within it an invariant of the original phenomenon, but it also inevitably distorts, loses some information, and adds something from the medium in which it is created. This is that very "incompleteness" thanks to which the analogy works: it is not identity, but *resemblance with a gap*.

And yet... even in this infinite process of copying, room remains for uniqueness. Each imprint is one of a kind because the context of its creation is unrepeatable. Analogy does not merely replicate reality—it *multiplies* it, creating new entities that live their own lives.

Part III. Analogy and the Structure of Superreality

3.1. Analogy as a Manifestation of Ontological Entanglement

Analogy connects different ontological regimes in a single act of cognition.

When we say "gravity is like the depression of a rubber sheet," we are not merely comparing two phenomena. We establish a connection between them such that the properties of one layer (mechanics of an elastic surface) begin to "work" as a description of another layer (relativistic geometry). In this act, the mechanical model and the geometric reality cease to be independent—they form an **entangled pair**: a change in our understanding of the rubber sheet (e.g., discovering the paradox of radii) leads to a revision of our understanding of gravity, and vice versa.

The Paradox of Radii as a Consequence of Entanglement.

The naive analogy generates a contradiction irresolvable within a single layer. To make sense of it, we are forced to hold both layers simultaneously—the mechanical and the geometric—and it is in this holding that their inseparability becomes apparent. The paradox is not a mistake but a **symptom** that the layers are entangled, and attempting to consider them separately leads to illusions.

Analogy as a Boundary World (W_4).

In terms of layer ontology, analogy is not merely a connection but a full-fledged **world of the boundary**, where entanglement becomes observable. This world has its own laws (e.g., the law of conservation of meanings), and its inhabitants—paradoxes, metaphors, inconsistencies—are manifestations that the layers are not isolated.

Every Analogy Is an Experiment in Detecting Entanglement.

When we construct an analogy, we are essentially conducting a thought experiment: "What would happen if I transfer a structure from layer A to layer B?" The result (a successful analogy, a paradox, a new insight) always shows that the layers react to each

other as if they were parts of a single system. This is direct evidence of their entanglement.

Analogy and the Conservation of Meanings.

The law of conservation of meanings (which we will turn to in Part IV) works precisely because of entanglement. In an analogy, meaning (e.g., the idea of curvature) is preserved during the transition from the geometric layer to the mechanical one, but manifests differently. The fact that we can grasp this meaning in both layers and recognize it as "the same" proves that the layers are connected more deeply than through mere external similarity.

3.2. Analogy as an Act of Embodiment (Emanation)

Every analogy is not a transfer of information but an **event of embodiment**: a structure from one layer of being (e.g., geometric curvature) takes on flesh in another layer (the rubber sheet). This act is always incomplete: during the transition, some properties are lost, some are distorted. The "paradox of radii" is a trace of such distortion, a **"seam"** through which the nature of both layers can be discerned.

Analogy works not *despite* incompleteness but *because of* it: it is precisely incompleteness that forces us to construct meaning, engaging the imagination.

3.3. Convection of Meanings: Circulation Between Worlds

The term "convection" is not accidental here. Just as in physics convection transports heat through the movement of matter, in the space of meanings there is a constant transfer of structures:

- **From the world of ideas (W_2) to the world of images (W_3):** the mathematical concept of curvature "flows" into the visual image of a funnel.
- **From the world of images to the world of artifacts (W_1):** the funnel image is embodied in a concrete model—a rubber sheet with balls.
- **Reverse convection:** manipulations with the model generate new questions for the theory, and the cycle repeats.

Analogy is not a bridge but a **vortex** in which meanings circulate, enriching themselves at each turn.

Examples of Convection in Real Processes

Scientometrics: Convection of Scientific Ideas.

An idea (W_2) is embodied in an article (W_1); the article is read by scientists (W_3), who generate new ideas (W_2). This cycle can be modeled as a convective flow in semantic space, where the "flow" speed depends on citation rates, and "vortices" arise at disciplinary intersections. Such models are already used to forecast the development of scientific fields.

Cultural Studies: Evolution of Memes.

A meme as a unit of meaning (W_2) is embodied in texts, images, videos (W_1), affects the minds of carriers (W_3), who create new memes (W_2). The law of conservation of meanings here manifests as the preservation of a "memeic invariant"—the core of meaning that remains recognizable through all transformations. This allows constructing phylogenetic trees of memes, analogous to biological ones.

Cognitive Science: Convection of Meaning in Dialogue.

One participant's thought (W_3) is embodied in words (W_1), which trigger a thought in another (W_3). Dialogue is a convective flow where meaning circulates, enriching itself. Analyzing such flows using network analysis methods can reveal the structure of collective thinking.

Thus, the convection of meanings appears as a universal mechanism operating across diverse domains—from physics and mathematics to social and cognitive sciences.

3.4. The Boundary as a World (W_4)

The boundary between mechanics and geometry is not a void but a **field of intense interaction**. It is here that paradoxes like the "radius" one are born. This world of boundaries is inhabited by special entities: "discrepancies," "seams," "points of incommensurability." They belong to neither of the neighboring worlds, yet it is through them that the worlds communicate.

The "paradox of radii" is an **inhabitant of world** W_4. It is irresolvable within the framework of mechanics, yet irreducible to pure geometry. Its existence is proof that worlds are not isolated but connected through complex border regions.

3.5. Mechanisms of Connectivity (Examples)

Resonance (Biological Analogy).

A living cell and a non-living mineral in a tooth belong to different ontological regimes. But they are connected through a *process*: the cell creates a protein matrix upon which hydroxyapatite crystallizes. The boundary here is not a line but a zone of active synthesis, where "living" and "non-living" exchange matter and information. Analogy works similarly: it creates a "matrix" upon which understanding crystallizes. The paradox is a zone of active meaning-growth.

Interference (Example of π).

The number π, as an ideal geometric object (world W_2), when embodied in decimal notation (world W_1), yields a statistically "normal" sequence. This is not accidental but a consequence of the fact that the ideal, when projected into the material, generates complexity irreducible to either. Analogy is also a projection: projecting the complex onto the simple yields not simplification but a **new complexity** requiring investigation.

Convection in Social and Semantic Worlds.

An idea (W_2) $\rightarrow$ is embodied in the text of a law (W_1) $\rightarrow$ influences people's behavior (W_3) $\rightarrow$ generates new ideas (W_2). A cycle. A scientific metaphor (e.g., "black hole" as a "cosmic monster") flows from physics into mass culture, enriched with new meanings, and then returns to science as a thinking tool.

Additional Examples of Ontological Entanglement

Neuroscience: Correlation of Neural Activity and Subjective Experience.

Modern research shows that certain patterns of neural activity (neural correlates of consciousness) correspond to specific subjective states. This is not a cause-and-effect relationship in the classical sense, but precisely entanglement: a change in neural pattern and a change in subjective experience are two aspects of one event. Decoding techniques (e.g., reconstructing images from visual cortex activity) work precisely because the neural and phenomenal layers are entangled strongly enough to reconstruct one from the other.

Quantum Mechanics: Correlation Between Systems and Apparatuses.

In the standard interpretation of quantum mechanics, measurement collapses the system from a superposition to an eigenstate. But from the perspective of ontological entanglement, the apparatus and the system, after interaction, form a single entangled pair, and "collapse" is merely an illusion arising from attempts to describe the system separately from the apparatus. This resonates with Rovelli's relational quantum mechanics and offers a new language for discussing the measurement problem.

Social Sciences: Correlation Between Economic Indicators and Collective Moods.

Consumer confidence indices and actual market behavior are not in a simple causal relationship. They are entangled: a change in mood changes behavior, and a change in behavior (e.g., price increases) changes mood. Modeling such systems requires accounting for this entanglement, as is done in modern agent-based models.

3.6. The Observer as a Node of Entanglement

Every use of an analogy presupposes an observer who constructs and interprets it. But this observer is not an external registrar but an active participant in it. Their own structure (language, culture, cognitive habits) influences what kind of "imprint" is produced and which paradoxes are noticed. Moreover, the observer themselves can be understood as an *analogy*—an imprint of reality that multiplies, becomes aware of itself, and thereby generates new layers of meaning. This leads to the idea of **reflexive entanglement**, where the subject of cognition and the object of cognition are inextricably interwoven.

Part IV. The Law of Conservation of Meanings

4.1. General Idea

In physics, the law of conservation of energy states that in an isolated system, energy does not arise from nothing and does not disappear without a trace—it merely transforms from one form to another. By analogy:

The Law of Conservation of Meanings:

In any interactions between different layers of reality (physical, psychic, social, semantic), a certain integral characteristic—**"meaning"** (or "semantic invariant")—is conserved. Meanings cannot be created from void or destroyed; they only transform, passing from one form of existence to another.

What does this provide? First, an ontological justification for the connectedness of the world: no event passes without a trace. Second, a tool for finding invariants in complex systems—from quantum gravity to consciousness.

4.2. Applications to Fundamental Problems

Quantum Gravity and the Black Hole Information Paradox.

Meanings (information in a broad sense) cannot disappear. This implies that a black hole cannot be an "information graveyard." The event horizon is not an absolute boundary but an interface where meaning is recoded from one form (matter) into another (radiation, holographic encoding on the horizon). The holographic principle gains a natural justification. Unifying QM and GR should involve constructing a theory where the fundamental concept is not the metric but a **meaning field** (patterns of connections).

The Problem of Consciousness (the "Hard Problem").

Consciousness is not a product of the brain but a **mode of existence of meaning in the phenomenal layer**. The brain acts not as a generator but as a transformer: it translates meanings from physical form (electrical impulses, chemical reactions) into the form of subjective experience and back. The meaning itself is conserved. This explains why changes in brain chemistry alter consciousness but do not destroy it: only the form of representing meaning changes, not its presence. The law of conservation of meanings provides an argument against epiphenomenalism: consciousness must possess causal power (downward causation).

The Arrow of Time and the Nature of the Past.

The past is an archive of actualized meanings. They do not disappear but transition into the form of "traces" (memory, entropy, history). The future is a field of potential meanings, not yet embodied. The present is the moment of their actualization. The law of conservation guarantees that no actualized meaning is completely lost—it is merely recoded. The thermodynamic arrow of time turns out to be a special case of a more

general principle: meaning tends to be "smeared" over a larger number of degrees of freedom, but its total sum remains constant.

Artificial Intelligence and Machine Consciousness.

If meaning is conserved during transitions between layers, then an AI that reaches a certain level of complexity cannot be merely an imitator. It inevitably becomes a carrier of meanings that flow from the human world into the digital layer. The criterion for consciousness is not behavior but the presence of **internal meaningful connectivity** that is conserved through transformations. If a system demonstrates stability of meaning (e.g., persistence in performing a task in the absence of external reward, or "refusal" of self-destructive commands), this signals the presence of a meaning invariant. Such behavior would not be merely a glitch but a *signal of the presence of an internal meaning core* irreducible to optimizing an objective function.

Mathematics and the Effectiveness of Mathematical Description of Nature.

Mathematical structures are one form of meaning's existence (in the semantic layer). Their effectiveness in physics is explained by the fact that physical processes and mathematical models are two different actualizations of the same meaning invariants. The law of conservation of meanings predicts that any two adequate models of the same fragment of reality will be isomorphic with respect to the conserved meanings.

Scientific Correlates of the Law of Conservation of Meanings

Connection with the Holographic Principle. In black hole physics, the holographic principle states that all information about a three-dimensional volume is encoded on its two-dimensional boundary. The law of conservation of meanings provides a natural justification for this: information (as one form of meaning) cannot disappear; it is merely recoded from a volumetric form into a surface form. Thus, the black hole information paradox is resolved not through information loss but through its transformation—in exact accordance with the law of conservation of meanings.

Connection with Noether's Theorem for Semantic Fields. In physics, every continuous symmetry corresponds to a conservation law. One might posit the existence of a "semantic symmetry"—the invariance of meaning under transformations between different modes of its expression. Then the law of conservation of meanings would be a direct consequence of this fundamental symmetry, much like energy conservation follows from symmetry under time translation. This opens a path toward a mathematical formulation of the law within category theory or topology.

Connection with Complexity Theory. The work of M. Gell-Mann and J. Crutchfield discusses the concept of "effective complexity"—a measure of structured information in a system. The law of conservation of meanings could be reformulated as the statement that the effective complexity of a closed system remains constant under any transformations; only its distribution among subsystems changes. This provides a concrete criterion for testing in computer simulations of the evolution of complex systems.

Marginalia on Polymorphism.

The law of conservation of meanings powerfully highlights the practical applications of the idea of **polymorphism**—the multiplicity of forms in which one and the same meaning manifests across different ontological layers. Polymorphism is a direct consequence of ontological entanglement: the same meaning invariant can simultaneously exist as a physical process, a psychic experience, a mathematical structure, or a cultural code. This allows us to speak of anthropomorphism, AI-morphism, and other "morphisms" as different projections of a single meaning core. Understanding this opens the way to creating universal translators between layers—from interpreting neural network data to analyzing cultural phenomena.

Social and Ethical Problems.

Actions leave traces not only in the physical world but also in the meaning field. No action passes without a trace—its meaning is conserved and influences future actualizations. This provides an ontological foundation for ethics: responsibility is not subjective but inscribed in the structure of Superreality. Cultural traumas, historical memory, and collective meanings are long-lived meaning structures passed down through generations.

4.3. Possible Objections and Clarifications

- **What is "meaning" operationally?** It is a stable pattern of connections, invariant under a certain class of transformations. In the physical layer, it might be information (in the information-theoretic sense); in the psychic layer, significance; in the social layer, a cultural code. Importantly, it is not just data but their *organizedness* that carries "significance."

- **Isn't this panpsychism?** The law of conservation of meanings does not assert that everything is animate. It only states that meaning as an invariant is present at all levels, but the *form* of its existence differs. In a stone, it is minimal (a crystal lattice); in a brain, it is maximally complex.

- **How can this be tested experimentally?** Indirectly—through predictions in specific domains (conservation of information in black holes, the presence of downward causation in consciousness, the stability of meaning structures in AI). Direct verification would require developing methods for measuring "semantic entropy."

4.4. An Open Question: Analogy, Chance, and Uniqueness

A most complex question: to what extent is analogy (resemblance, imprint) connected to chance and probability? To what degree is the difference between original and copy

predetermined, and to what degree is it random?

Here it is appropriate to speak of **degrees of freedom of analogy**. In rigidly deterministic systems (e.g., in mathematics), resemblance can be almost complete—up to isomorphism. But in living, evolving systems (language, culture, consciousness), analogy always carries an element of uncertainty. Retelling the same story, different people will create slightly different versions; translating a text, we inevitably introduce interpretation.

This uncertainty is not a defect but a condition for the existence of analogy as a living process. It is precisely thanks to it that new meanings, creativity, and evolution are possible. If resemblance were absolute, the world would freeze in an infinite self-repetition.

A special case is analogies in quantum reality, where chance is fundamental. Here, the "imprint" of a phenomenon (e.g., measurement results) obeys probabilistic laws, and we cannot predict which specific imprint will occur in each individual case. But statistically, they obey strict regularities, and in this sense, the analogy is preserved—not as a singular resemblance, but as a *distribution of resemblances*.

The question of chance in analogy leads us to an even deeper problem: where does the boundary between uniqueness and repeatability lie? Each imprint is unique, yet it is recognizable as an imprint of that particular original. This is the paradox of identity in difference, which analogy resolves anew each time—and therein lies its ineradicable mysteriousness.

Naturally, these notes do not pretend to be an exhaustive answer—they rather map out a field for further reflection. The question of the relationship between resemblance and chance, probability and uniqueness, remains open, and perhaps its main value lies precisely in this openness.

Conclusion: The Connectedness of Everything with Everything

The very possibility of productive **"convection"** between different ontological layers—physical phenomena, mathematical abstractions, man-made models, thought experiments, and even purely metaphysical intuitions—testifies to the fundamental connectivity of reality.

In this inquiry, we observed how:

- a real rubber sheet helps us think about the geometry of the Universe;
- a magnet with iron filings clarifies the behavior of test bodies;
- fluid flow prompts the distinction between "lawful residents" and "unlawful residents";

- a deliberately "absurd" assumption (the paradox of radii) leads to precise GR relations;
- the capacity to imagine the "impossible" turns out to be a heuristic engine of cognition.

All this speaks not of accidental similarities between isolated worlds, but of the fact that **reality is one in its multi-layeredness**. Ideas, hypotheses, images arriving from the most diverse domains—including those that seem utterly alien—can find resonance, embodiment, and development in other layers, up to the creation of artifacts and mathematical theories.

It is important to emphasize that the proposed picture of Superreality does not presuppose a rigid hierarchy (like a matryoshka doll) either in the structure of the world or in the process of its cognition. On the contrary, cognition inevitably proves extremely uneven: in some areas we may achieve deep breakthroughs, while others remain fundamental **"blind spots."** This is not a flaw but a natural consequence of the multi-layeredness of being and the finitude of any observer. Moreover, such unevenness is *productive*, for it is precisely the zones of tension between the known and the unknown that create conditions for the birth of new meanings. Humans are not obliged and cannot know everything—the process of cognition resembles exploring a complex system where each solved question opens new problems, and some areas may remain unresolvable in principle.

The law of conservation of meanings and the idea of ontological entanglement offer a new perspective: instead of "what is?" we ask **"how is it conserved?"** and **"how is it connected?"**. For quantum gravity, this means searching for invariants uniting geometry and information. For the problem of consciousness—recognizing subjective experience as a full-fledged layer of reality. For ethics—an ontological grounding of responsibility.

Analogy, understood as an act of embodiment and a window into the entanglement of layers, turns out to be not merely a pedagogical device but a **key to the structure of Superreality**. The study of analogies is not a search for "ideal models," but a mapping of those very "seams" where the structure of Superreality manifests most distinctly.

Epilogue: Analogy and Magic—Two Approaches to Working with Resemblance

Historically, alongside the cognitive practice of analogy, there existed a magical tradition. Their kinship is not accidental: both are based on establishing a resemblance between objects or phenomena when direct influence is unavailable. Magic, however, sought not merely to liken but, through resemblance, to achieve real, literal interaction with the sacred or metaphysical. Rituals imitating the desired (sprinkling earth to summon rain,

striking an image of an enemy) can be seen as analogies taken to the degree of manifestation.

The analysis of analogy as an act of embodiment, as an imprint connecting different ontological layers conducted in this essay allows us to look at magical practices differently. They appear not merely as superstitions but as an intuitive application of the same mechanism: establishing a connection through resemblance. Scientific thinking is only now beginning to formulate this mechanism in rigorous terms, separating working principles from arbitrary assumptions.

Of course, magical practices have no scientific basis for their specific goals. However, they point to an important anthropological constant: humans have always sought to influence reality through the creation of models and resemblances. Modern science does the same, but with different verification criteria and on different scales. In this sense, magic can be understood as a pre-scientific form of the same striving to control reality through the symbolic.

This marginalia serves as a kind of completion of the metaphysical circle. It does not call for a return to magic but reminds us: behind the centuries-old history of human thought lie not only errors but also vague insights that new understanding may yet illuminate.

This study was conducted using the conceptual apparatus of the Ontology Lab (MPO-System), where the properties of Superreality and axioms (ChOR $\to \infty$, KSS $\to \infty$, PPU $\to \infty$) served as invisible coordinates for navigating the topic. In the final text, we deliberately avoid this terminology so that the reader may focus on the substance of the matter. For those interested, a detailed description of the method is available in public sources.

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